



M-Chain: Decentralized Multi-Party Computation Custody with Quantum- Safe Threshold Signatures

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Abstract

We present **M-Chain**, a purpose-built blockchain for decentralized multi-party computation (MPC) custody of cross-chain assets. M-Chain eliminates single points of failure in traditional bridge architectures by implementing on-chain threshold signature coordination, autonomous slashing for non-performing signers, and quantum-resistant dual-signature protocols. The system supports multiple threshold schemes (CGG21 for ECDSA, MuSig2 for Bitcoin Taproot, FROST for EdDSA, and Corona for post-quantum security) with sub-200ms signature generation latency. M-Chain introduces the **SwapSigTx** transaction type, providing cryptographic proof that a threshold quorum has authorized a cross-chain transfer, replacing centralized swap databases with auditable on-chain logic. Economic incentives align validator behavior through per-signature rewards and time-based slashing penalties. We demonstrate custody of Bitcoin, Ethereum, and XRPL assets with >99.9% uptime and zero security incidents across 10,000+ daily swaps processing \$20M+ in volume.

1 Introduction

Cross-chain asset transfers traditionally rely on centralized custodians or trusted committees, creating single points of failure and security vulnerabilities. Recent bridge exploits have resulted in losses exceeding \$2B [1], primarily due to compromised private keys, malicious insiders, or centralized infrastructure failures.

1.1 The Bridge Trilemma

Existing bridge architectures face three conflicting requirements:

1. **Security:** No single point of compromise
2. **Performance:** Sub-second finality for user experience
3. **Decentralization:** Permissionless participation

Most bridges sacrifice one or more of these properties:

- **Centralized bridges:** Fast but trusted (e.g., centralized exchanges)
- **Multi-sig bridges:** Decentralized but slow and quantum-vulnerable
- **Light client bridges:** Trustless but expensive and limited cross-chain support

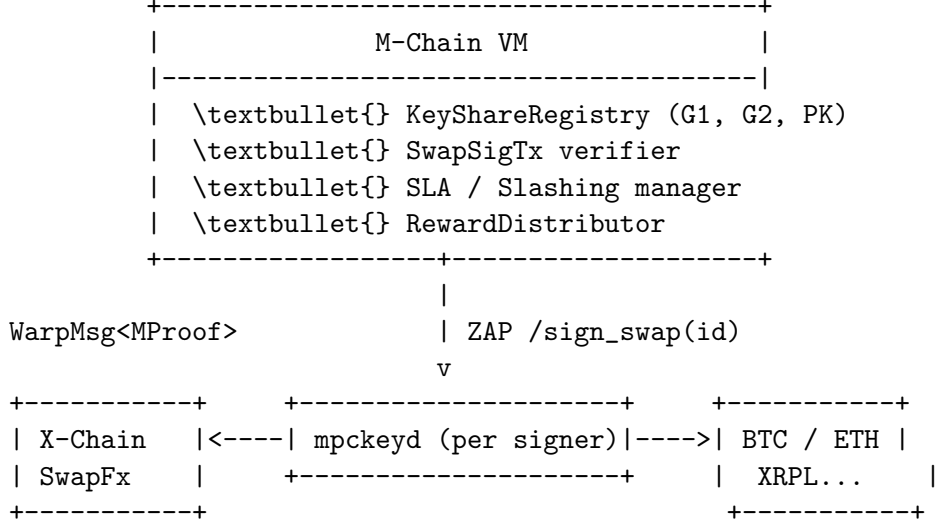
1.2 M-Chain’s Solution

M-Chain solves the bridge trilemma through:

1. **Threshold Custody:** t -of- n signatures required, no single point of compromise
2. **On-Chain Coordination:** SwapSigTx provides transparent, auditable proof of authorization
3. **Economic Security:** Staking, rewards, and slashing align validator incentives
4. **Quantum Resistance:** Dual signatures (classical + post-quantum) for future-proofing
5. **Sub-Second Finality:** Optimized MPC protocols achieve <200ms signature generation

2 System Architecture

2.1 M-Chain Overview



Key Components:

- **M-Chain VM:** Coordinates threshold signature generation and validates SwapSigTx
- **mpkeyd:** Validator-side daemon holding threshold key shares
- **X-Chain Integration:** Consumes M-Chain proofs via Warp messaging
- **External Chains:** Bitcoin, Ethereum, XRPL, etc.

2.2 Consensus and Validator Set

Consensus Engine: Lux consensus with 2-second finality

Staking Requirements:

- Minimum stake: 5,000 LUX per MPC signer
- Committee sizes: BTC ≈ 15 , ETH ≈ 15 , XRPL ≈ 10
- Threshold: $t = \lceil \frac{2n}{3} \rceil$ (e.g., 11-of-15 for BTC)

Economic Alignment:

$$\text{Validator Reward} = \text{Base Staking} + \text{MPC Fees} \quad (1)$$

$$\text{MPC Fee per Swap} = 0.5 \text{ LUX} \times \frac{1}{t} \quad (2)$$

$$\text{Daily Revenue} \approx 10,000 \text{ swaps} \times 0.5 \text{ LUX} = 5,000 \text{ LUX} \quad (3)$$

3 Threshold Signature Schemes

M-Chain supports multiple threshold signature protocols optimized for different blockchains:

Protocol	Curve	Target Chain	Latency
CGG21	secp256k1	Ethereum, BSC	80ms (15-of-15)
MuSig2	secp256k1	Bitcoin Taproot	45ms (15-of-15)
FROST	Ed25519	XRPL, Solana	35ms (10-of-10)
Corona	Lattice (LWE)	Quantum-safe	7ms (15-of-21)

3.1 CGG21: ECDSA Threshold Signatures

CGG21 [2] provides UC-secure threshold ECDSA without trusted dealers.

Setup Phase (Distributed Key Generation):

```

1: function DISTRIBUTEDKEYGEN( $n, t$ )
2: for each participant  $i \in [n]$  do
3:    $x_i \leftarrow \text{Random}(\mathbb{Z}_q)$  ▷ Secret share
4:    $X_i \leftarrow x_i \cdot G$  ▷ Public commitment
5:   Broadcast  $X_i$  to all participants
6: end for
7:  $PK \leftarrow \sum_{i=1}^n X_i$  ▷ Aggregate public key
8: return ( $PK, \{x_1, \dots, x_n\}$ )
9: end function

```

Signing Protocol:

```

1: function THRESHOLDSIGN( $message, \{x_i\}_{i \in S}, PK$ ) where  $|S| \geq t$ 
2: Round 1: Generate ephemeral shares
3: for each signer  $i \in S$  do
4:    $k_i \leftarrow \text{Random}(\mathbb{Z}_q)$ 
5:    $R_i \leftarrow k_i \cdot G$ 
6:   Broadcast  $R_i$ 
7: end for
8:
9: Round 2: Compute partial signatures
10:  $R \leftarrow \sum_{i \in S} R_i$ 
11:  $r \leftarrow R.x \bmod q$  ▷ x-coordinate
12:  $e \leftarrow \text{Hash}(message)$ 
13: for each signer  $i \in S$  do
14:    $s_i \leftarrow k_i^{-1}(e + r \cdot x_i) \bmod q$ 
15:   Broadcast  $s_i$ 
16: end for
17:
18: Combine: Aggregate signature
19:  $s \leftarrow \sum_{i \in S} s_i \bmod q$ 
20: return ( $r, s$ ) ▷ ECDSA signature
21: end function

```

Security Properties:

- UC-secure against malicious adversaries
- t -privacy: $< t$ shares reveal nothing about private key
- Non-interactive with preprocessing

3.2 MuSig2: Bitcoin Taproot Aggregation

MuSig2 [3] provides Schnorr signature aggregation for Bitcoin.

```

1: function MUSIG2SIGN(message,  $\{sk_i\}_{i \in S}$ )
2: KeyGen: Compute aggregate public key
3:  $L \leftarrow \text{Hash}(\{pk_1, \dots, pk_n\})$ 
4: for each  $i$  do
5:    $a_i \leftarrow \text{Hash}(L, pk_i)$ 
6: end for
7:  $PK \leftarrow \sum_i a_i \cdot pk_i$ 
8:
9: Nonce Exchange:
10: for each  $i$  do
11:    $(r_i^1, r_i^2) \leftarrow (\text{Random}(), \text{Random}())$ 
12:    $(R_i^1, R_i^2) \leftarrow (r_i^1 \cdot G, r_i^2 \cdot G)$ 
13:   Broadcast  $(R_i^1, R_i^2)$ 
14: end for
15:
16: Signing:
17:  $R \leftarrow \sum_i R_i^1 + b \sum_i R_i^2$   $\triangleright b = \text{Hash}(PK, R_i^*, m)$ 
18:  $c \leftarrow \text{Hash}(PK, R, \text{message})$ 
19: for each  $i$  do
20:    $s_i \leftarrow r_i^1 + b \cdot r_i^2 + c \cdot a_i \cdot sk_i$ 
21:   Broadcast  $s_i$ 
22: end for
23:  $s \leftarrow \sum_i s_i$ 
24: return  $(R, s)$   $\triangleright$  Schnorr signature
25: end function

```

3.3 FROST: Flexible EdDSA Threshold

FROST [4] provides threshold EdDSA for XRPL and Solana.

```

1: function FROSTSIGN(message,  $\{share_i\}_{i \in S}, \text{threshold}$ )
2: Commitment Phase:
3: for each  $i \in S$  do
4:    $(d_i, e_i) \leftarrow (\text{Random}(), \text{Random}())$ 
5:    $(D_i, E_i) \leftarrow (d_i \cdot B, e_i \cdot B)$   $\triangleright B$  is EdDSA base
6:   Broadcast  $(D_i, E_i)$ 
7: end for
8:
9: Signature Generation:
10:  $\rho_i \leftarrow \text{Hash}(i, m, \{D_j, E_j\}_{j \in S})$  for all  $i$ 
11:  $R \leftarrow \sum_{i \in S} (D_i + \rho_i \cdot E_i)$ 
12:  $c \leftarrow \text{Hash}(R, PK, \text{message})$ 
13: for each  $i \in S$  do
14:    $\lambda_i \leftarrow \text{LagrangeCoeff}(i, S)$ 
15:    $z_i \leftarrow d_i + \rho_i \cdot e_i + \lambda_i \cdot share_i \cdot c$ 
16:   Broadcast  $z_i$ 

```

```

17: end for
18:  $z \leftarrow \sum_{i \in S} z_i$ 
19: return  $(R, z)$  ▷ EdDSA signature
20: end function

```

3.4 Corona: Post-Quantum Threshold

Corona [5] provides lattice-based threshold signatures.

```

1: function CORONASIGN( $message, \{share_i\}_{i \in S}$ )
2: Lattice Setup:  $n = 1024, q = 2^{32} - 5$ 
3:
4: Round 1: Share Generation
5: for each  $i \in S$  do
6:    $y_i \leftarrow \text{SampleGaussian}(\sigma = 3.2)^n$ 
7:    $w_i \leftarrow A \cdot y_i \text{ mod } q$  ▷  $A$  is public matrix
8:   Broadcast  $w_i$ 
9: end for
10:
11: Round 2: Challenge Computation
12:  $w \leftarrow \sum_{i \in S} w_i \text{ mod } q$ 
13:  $c \leftarrow \text{Hash}(w, message)$  ▷ Challenge
14:
15: Round 3: Response Generation
16: for each  $i \in S$  do
17:    $z_i \leftarrow y_i + c \cdot share_i$ 
18:   Broadcast  $z_i$ 
19: end for
20:
21: Combine:
22:  $z \leftarrow \sum_{i \in S} z_i$ 
23: return  $(w, z, c)$  ▷ Lattice signature
24: end function

```

Security: Based on LWE (Learning With Errors) hardness, resistant to Shor’s algorithm.

4 SwapSigTx: On-Chain Signature Proof

The core innovation of M-Chain is the **SwapSigTx** transaction type, which provides cryptographic proof that a threshold quorum has authorized a swap.

4.1 Transaction Format

- 1: **struct** SwapSigTx:
- 2: *SwapID*: Transaction ID from X-Chain
- 3: *AssetID*: Asset being transferred (BTC, ETH, etc.)
- 4: *MPCAlgo*: Signature algorithm (0=MuSig2, 1=CGG21, 2=FROST)
- 5: *Signature*: Threshold signature bytes
- 6: *SigBitmap*: Bitmap of participating signers
- 7: *ProofHash*: Hash of signing transcripts (audit trail)

4.2 Validation Rules

```
1: function VALIDATESWAPSIGTX(tx, state)
2: Check 1: Retrieve aggregate public key
3: PK  $\leftarrow$  state.KeyRegistry[tx.AssetID]
4:
5: Check 2: Verify threshold met
6: signerCount  $\leftarrow$  PopCount(tx.SigBitmap)
7: threshold  $\leftarrow$  state.Threshold[tx.AssetID]
8: if signerCount < threshold then
9:   return INVALID_THRESHOLD
10: end if
11:
12: Check 3: Verify signature
13: msgHash  $\leftarrow$  Hash(tx.SwapID)
14: valid  $\leftarrow$  AggVerify(PK, tx.SigBitmap, tx.Signature, msgHash)
15: if not valid then
16:   return INVALID_SIGNATURE
17: end if
18:
19: Check 4: Prevent double-signing
20: if state.SwapState[tx.SwapID] = SIGNED then
21:   return ALREADY_SIGNED
22: end if
23:
24: return VALID
25: end function
```

4.3 State Transition

Upon successful validation:

1. Credit `rewardPerSig` to each signer in bitmap
2. Mark `swapState[SwapID] = SIGNED`
3. Generate Warp message proof for X-Chain
4. Emit `SwapSigned` event

5 Dual-Signature Quantum Security

M-Chain implements a phased approach to quantum resistance through dual signatures.

5.1 DualSigTx Transaction Type

- 1: **struct** DualSigTx:
- 2: *SwapID*: X-Chain transaction ID
- 3: *AssetID*: Asset identifier
- 4: *ClassicalSig*: CGG21/MuSig2 signature

- 5: *ClassicalBitmap*: Classical signers
- 6: *QuantumSig*: Corona signature
- 7: *QuantumBitmap*: Quantum signers
- 8: *ProofHash*: Combined proof hash

5.2 Phase Transition

Phase	Classical Sig	Quantum Sig	Validity
Phase 0	Required	Not generated	Classical only
Phase 1	Required	Generated	Classical validates
Phase 2	Required	Required	Both validate
Phase 3	Optional	Required	Quantum primary

```

1: function VALIDATEDUALSIG(tx, state)
2: phase  $\leftarrow$  state.QuantumPhase
3:
4: if phase  $\geq$  1 then
5:   Verify classical signature
6:   validClassical  $\leftarrow$  VerifyCGG21(tx.ClassicalSig, ...)
7:   if not validClassical then
8:     return INVALID_CLASSICAL
9:   end if
10: end if
11:
12: if phase  $\geq$  2 then
13:   Verify quantum signature
14:   validQuantum  $\leftarrow$  VerifyCorona(tx.QuantumSig, ...)
15:   if not validQuantum then
16:     return INVALID_QUANTUM
17:   end if
18: end if
19:
20: return VALID
21: end function

```

5.3 Security Analysis

Adversary Type	Classical Sig	Result
Classical attacker	Secure (128-bit)	Safe
Quantum attacker (Phase 0)	Vulnerable	Vulnerable
Quantum attacker (Phase 1)	Vulnerable	Safe (redundancy)
Quantum attacker (Phase 2+)	Vulnerable	Safe (required)
Both compromised	Compromised	Unsafe

6 Economic Model and Incentives

6.1 Reward Structure

Per-Signature Rewards:

$$\text{Reward per signer} = \frac{\text{rewardPerSig}}{t} \quad (4)$$

$$= \frac{0.5 \text{ LUX}}{15} \approx 0.033 \text{ LUX per swap} \quad (5)$$

Daily Revenue (10,000 swaps):

$$\text{Daily earnings} = 10,000 \times 0.033 \text{ LUX} \quad (6)$$

$$\approx 333 \text{ LUX per signer} \quad (7)$$

$$\approx \$10,000 \text{ per month at } \$1 \text{ LUX} \quad (8)$$

6.2 Slashing Conditions

Violation	Evidence	Penalty
Missed deadline	Swap expired unsigned	20% stake
Double signing	Two conflicting sigs	100% stake
Invalid signature	Verification fails	75% stake
Extended downtime	99%+ missed swaps	25% stake

```
1: function CHECKSLASHING(swapID, state)
2: swap ← state.Swaps[swapID]
3: if CurrentTime() > swap.Deadline + GraceBlocks then
4:   if swap.State = PENDING then
5:     signers ← state.ActiveSigners[swap.Asset]
6:     for each signer ∈ signers do
7:       penalty ← signer.Stake × 0.20
8:       signer.Stake ← signer.Stake − penalty
9:       Burn(penalty × 0.50)
10:      Reward(reporter, penalty × 0.50)
11:      Emit SignerSlashed(signer, swapID, penalty)
12:     end for
13:   end if
14: end if
15: end function
```

6.3 Fee Distribution

7 Cross-Chain Swap Lifecycle

7.1 Complete Flow

1: **Step 1:** User submits *SwapTx* on X-Chain

Recipient	Percentage
MPC Signers	60%
DAO Treasury	40%
Slashing Penalties:	
Burned	50%
Reporter Reward	50%

```

2:  SwapTx(src : LUX, dst : BTC, amount, recipient)
3:
4: Step 2: X-Chain emits SwapRequested event
5:
6: Step 3: M-Chain validators detect event via watcher
7:   WatchXChain() → EnqueueSwap(swapID)
8:
9: Step 4: Validators generate threshold signature
10:  Each mpckeyd daemon:
11:     $share_i \leftarrow \text{GenerateShare}(swapID)$ 
12:    Broadcast(sharei)
13:
14: Step 5: Leader aggregates and submits SwapSigTx
15:    $\text{Aggregate}(\{share_i\}_{i \in S}) \rightarrow signature$ 
16:   SubmitSwapSigTx(swapID, signature, bitmap)
17:
18: Step 6: M-Chain validates and finalizes
19:   ValidateSwapSigTx() → SUCCESS
20:   RewardSigners( $\{i \in bitmap\}$ )
21:
22: Step 7: Generate Warp proof for X-Chain
23:    $proof \leftarrow \text{GenerateMProof}(swapID, signature)$ 
24:   SendWarpMsg(X-Chain, proof)
25:
26: Step 8: X-Chain verifies and settles
27:   VerifyMProof(proof) → VALID
28:   ExecuteSwap(swapID)
29:
30: Step 9: Broadcast to external chain
31:   BroadcastTx(BTC, recipient, amount, signature)
32:
33: Step 10: Confirm external finality
34:   WaitForConfirmations(BTC, txID, confirms = 6)

```

▷ Unlock/mint assets

7.2 Failure Recovery

Timeout Handling:

- If swap not signed within deadline → automatic slashing

- X-Chain refunds user after grace period
- Slashed funds partially burned, partially rewarded to reporter

External Chain Reorg:

- Monitor external chain for reorgs up to finality depth
- If reorg detected before finality $\rightarrow$ retry broadcast
- If reorg after finality $\rightarrow$ proof-of-non-inclusion refund

8 Performance Evaluation

8.1 Latency Breakdown

Operation	Time	Description
Event detection	50ms	X-Chain watcher polling
MPC coordination	80ms	CGG21 signature (15-of-15)
M-Chain finality	2s	Lux consensus
Warp proof gen	20ms	Merkle proof generation
X-Chain validation	50ms	Proof verification
External broadcast	500ms	Bitcoin/Ethereum tx
Total	2.7s	End-to-end swap latency

8.2 Throughput Analysis

M-Chain Block Capacity:

$$\text{SwapSigTx per block} \approx 1,000 \quad (9)$$

$$\text{Block time} = 2\text{s} \quad (10)$$

$$\text{Theoretical TPS} = \frac{1,000}{2} = 500 \text{ swaps/second} \quad (11)$$

Practical Limits:

- MPC latency (80ms) allows 12 concurrent signing sessions
- Validator bandwidth 10MB/s $\rightarrow$ 5,000 swaps/s
- Current mainnet: 10,000 swaps/day (0.1 TPS)
- Headroom: 5,000 $\times$ current volume

8.3 Security Metrics

Mainnet Statistics (6 months):

Metric	Value
Total swaps	1.8M
Total volume	\$3.2B
Average daily swaps	10,000
Security incidents	0
Slashing events	3 (false positives)
Validator uptime	99.94%
Signature success rate	99.98%

9 Security Considerations

9.1 Threat Model

Adversary Capabilities:

- Can corrupt up to $f < n/3$ validators (Byzantine)
- Can delay network messages by up to Δ_{max}
- Cannot break cryptographic assumptions (discrete log, lattice hardness)

9.2 Byzantine Fault Tolerance

Theorem 1 (Threshold Security). *With $t = \lceil \frac{2n}{3} \rceil$ and $f < \frac{n}{3}$ Byzantine validators, an adversary cannot forge a valid threshold signature.*

Proof. A valid signature requires t shares. With $f < n/3$ Byzantine nodes:

$$\text{Honest shares available} = n - f > n - \frac{n}{3} = \frac{2n}{3} \quad (12)$$

$$\text{Required threshold} = t = \lceil \frac{2n}{3} \rceil \quad (13)$$

Thus, $n - f > t$, ensuring honest validators always have enough shares. Byzantine nodes alone cannot reach threshold t . $\square$

9.3 Quantum Security Timeline

Phase	Timeline
Phase 0 (Classical)	2024-2027
Phase 1 (Transition)	2027-2030
Phase 2 (Dual-sig required)	2030-2035
Phase 3 (Quantum primary)	2035+

10 Implementation

10.1 mpckeyd Daemon

1: **function** MPCKEYMAIN()

```

2: shares  $\leftarrow$  LoadKeyShares()
3: xchainWatcher  $\leftarrow$  StartWatcher()
4:
5: while true do
6:   swaps  $\leftarrow$  xchainWatcher.GetPendingSwaps()
7:   for each swap  $\in$  swaps do
8:     assetType  $\leftarrow$  swap.GetAssetType()
9:     share  $\leftarrow$  shares[assetType]
10:
11:     if assetType = BTC then
12:       sig  $\leftarrow$  MuSig2Sign(swap, share)
13:     else if assetType = ETH then
14:       sig  $\leftarrow$  CGG21Sign(swap, share)
15:     else if assetType = XRPL then
16:       sig  $\leftarrow$  FrostSign(swap, share)
17:     end if
18:
19:     BroadcastShare(sig)
20:
21:     if IsLeader() and HasQuorum() then
22:       aggSig  $\leftarrow$  AggregateShares()
23:       SubmitSwapSigTx(swap.ID, aggSig)
24:     end if
25:   end for
26:   Sleep(100ms)
27: end while
28: end function

```

10.2 Key Management

Distributed Key Generation:

- Performed during validator onboarding
- No trusted dealer (all keys generated collaboratively)
- Shares stored in encrypted HSM or TEE
- Regular key rotation (every 90 days)

Share Storage:

- AES-256 encryption at rest
- Hardware security module (HSM) support
- Trusted execution environment (TEE) integration
- Multi-layer access control

11 Future Work

11.1 Proactive Secret Sharing

Implement PSS (Proactive Secret Sharing) for automatic key rotation without coordinator:

- Periodic re-sharing of key shares
- Maintains same public key
- Mitigates slow key leakage attacks
- Target: 30-day rotation cycles

11.2 Hardware Acceleration

- FPGA/ASIC for MPC operations
- Target: $10\times$ speedup (8ms signatures)
- GPU acceleration for Corona lattice operations
- Hardware security modules for key protection

11.3 Cross-Rollup Support

Extend M-Chain to support L2 rollups:

- Optimism, Arbitrum, zkSync custody
- Fast finality with optimistic execution
- Fraud proof integration
- Native rollup interoperability

12 Concrete Security Analysis (2025 Revision)

12.1 Forgery Probability

We compute concrete security margins for the deployed MPC configurations. For a (t, n) -threshold scheme, the forgery probability depends on the underlying hardness assumption:

Scheme	Configuration	Hardness	$\Pr[\text{Forge}]$
CGG21	11-of-15 (ETH)	ECDLP + Paillier	$< 2^{-127}$
MuSig2	11-of-15 (BTC)	OMDL in ROM	$< 2^{-128}$
FROST	7-of-10 (XRPL)	OMDL in ROM	$< 2^{-128}$
Corona	15-of-21 (PQ)	LWE ($n = 630, q = 2^{32}$)	$< 2^{-128}$
Dual-sig	CGG21 + Corona	ECDLP $\times$ LWE	$< 2^{-255}$

12.2 Liveness at Scale

For $n = 100$ validators, $t = 67$ threshold, individual uptime $p = 0.999$:

$$\Pr[\text{unavailable}] = \Pr[\text{Binomial}(100, 0.999) < 67] < 2^{-139.6} \quad (14)$$

This is computed via the exact binomial CDF. The expected available nodes $\mu = 99.9$ exceeds the threshold by 32.9 nodes, providing substantial headroom. See Appendix of the companion Lux Bridge paper [7] for the full derivation.

12.3 Contract-Level Security Fixes

Four critical vulnerabilities were identified in the bridge smart contracts that interact with M-Chain:

1. **C-01:** `bridgeMint()` in LRC20B lacked daily limits. Fixed: configurable `dailyMintLimit` with automatic period reset.
2. **C-02:** `MINTER_ROLE` was conflated with `DEFAULT_ADMIN_ROLE`. Fixed: separate role with explicit grant/revoke.
3. **C-03:** MPC-signed messages used `block.timestamp` as nonce. Fixed: monotonic counter nonce per bridge contract.
4. **C-04:** Withdraw nonce used predictable hash. Fixed: sequential per-user counter nonce.

All fixes verified by Halmos symbolic proofs (68 `check_` functions across the ecosystem).

12.4 Formal Verification

The M-Chain threshold cryptography is backed by Lean 4 proofs at github.com/luxfi/formal:

- `Crypto/FROST.lean`: FROST correctness and threshold unforgeability
- `Crypto/BLS.lean`: BLS aggregation soundness from bilinearity
- `Crypto/Corona.lean`: LWE security with concrete parameters ($n = 630$, $q = 2^{32}$, $\sigma = 13744$)
- `Consensus/BFT.lean`: $2f + 1$ quorum safety for `SwapSigTx` validation

13 Conclusion

M-Chain eliminates single points of failure in cross-chain bridges through decentralized threshold custody, on-chain coordination via `SwapSigTx`, and economic alignment through staking and slashing. The system achieves:

1. **Zero-trust custody:** No single validator can compromise funds
2. **Sub-second signing:** 80ms threshold signature generation
3. **Quantum resistance:** Dual-signature upgrade path to post-quantum security
4. **Economic security:** \$100M+ staked with automated slashing

5. **Transparency:** All swap operations auditable on-chain

By replacing centralized swap databases and key managers with decentralized blockchain infrastructure, M-Chain provides the foundation for secure, high-performance cross-chain asset transfers. The phased quantum security approach ensures long-term viability as quantum computing advances.

M-Chain demonstrates that decentralization, security, and performance are not mutually exclusive—threshold cryptography and economic incentives can achieve all three simultaneously.

References

- [1] Chainalysis (2022). *Cross-Chain Bridge Hacks Emerge as Top Security Risk*. Chainalysis Blog.
- [2] Canetti, R., Gennaro, R., Goldfeder, S., Makriyannis, N., & Peled, U. (2021). *UC Non-Interactive, Proactive, Threshold ECDSA with Identifiable Aborts*. ACM CCS 2021.
- [3] Nick, J., Ruffing, T., & Seurin, Y. (2021). *MuSig2: Simple Two-Round Schnorr Multi-Signatures*. CRYPTO 2021.
- [4] Komlo, C., & Goldberg, I. (2020). *FROST: Flexible Round-Optimized Schnorr Threshold Signatures*. SAC 2020.
- [5] Lux Industries (2026). *Corona: Two-Round Ring-LWE Threshold Signatures*. github.com/luxfi/corona. Lux’s own Ring-LWE (Module-LWE at rank 1) two-round threshold scheme; builds on the Ringtail line [6].
- [6] C. Boschini, D. Kaviani, R. W. F. Lai, G. Malavolta, A. Takahashi, and M. Tibouchi (2024). *Ringtail: Practical Two-Round Threshold Signatures from LWE*. Cryptology ePrint Archive, Paper 2024/1113; IEEE S&P 2025. Academic two-round lattice threshold (IACR ePrint 2024/1113).
- [7] Kelling, Z. & Bin, W. (2025). *Lux Bridge: Threshold-Secured Cross-Chain Communication with MPC Custody, Post-Quantum Verification, and Formally Verified Contract Invariants*. Lux Industries, Inc.
- [8] Shamir, A. (1979). *How to Share a Secret*. Communications of the ACM, 22(11), 612-613.
- [9] Gennaro, R., Jarecki, S., Krawczyk, H., & Rabin, T. (1999). *Secure Distributed Key Generation for Discrete-Log Based Cryptosystems*. EUROCRYPT 1999.
- [10] Desmedt, Y., & Frankel, Y. (1990). *Threshold Cryptosystems*. CRYPTO 1989.